

Introduction to Spatial processing based Filtering

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Definition: Spatial filtering refers to the process of modifying an image based on the values of neighboring pixels.

- It is typically done through convolution, where an image is convolved with a filter or kernel.
- **Purpose:** Image enhancement, noise reduction, edge detection, etc.

Spatial Domain vs Frequency Domain

Spatial Domain: Direct manipulation of pixel values.

Frequency Domain: Modification of the frequency components (using Fourier Transform).

Key Point: Spatial filtering operates directly on the pixels in an image.

Types of Filters

Low Pass Filters (LPF): These filters allow low-frequency components (smooth areas) to pass through and block high-frequency components (edges or noise).

High Pass Filters (HPF): These filters allow high-frequency components (edges, fine details) to pass through and block low-frequency components (smooth regions).

Low-pass Filter

Definition: Low Pass Filter (LPF) – Concept

- Purpose: To smooth the image or reduce noise by averaging neighboring pixels.

Characteristics:

- Removes high-frequency components (like noise and details).
- Produces a blurred or smoother version of the image.



High-pass Filter

Definition: High Pass Filter (HPF) – Concept

- Purpose: To emphasize edges or fine details by removing low-frequency components.

Characteristics:

- Highlights areas of rapid intensity change (edges).
- Can enhance sharp transitions and detect boundaries.



Common Low Pass Filters (LPF)

Definition:

- **Box Filter (Average Filter):** The simplest low pass filter, which averages neighboring pixels.
- **Gaussian Filter:** A weighted average filter that reduces high-frequency noise and blurring.

- Formula:
$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$$

Common High Pass Filters (HPF)

Definition:

Ideal High Pass Filter: It subtracts the low-frequency components from the image.

Laplacian Filter: Detects edges by looking for changes in intensity..

- Formula: $\nabla^2 I(x, y) = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$

Convolution Operation

Definition: The process of applying a filter (kernel) to an image through convolution.

- It involves multiplying the kernel values with the corresponding image pixel values and summing the results.
- **Mathematical Expression:**
$$g(x, y) = \sum_{i=-k}^k \sum_{j=-k}^k f(x + i, y + j) \cdot h(i, j)$$
 - $f(x, y)$: Input image
 - $h(i, j)$: Filter (kernel)

Correlation Operation

Definition: Correlation is a measure of similarity between two signals (images, in this case).

- It compares the pixel values of an image with a kernel (filter) and produces an output image based on this similarity.
- Purpose: Used for template matching, pattern recognition, and feature detection in image processing..

Convolution Vs Correlation

Definition:

- **Convolution:** A mathematical operation that combines an image with a filter (kernel) to create an output image. It involves flipping the kernel.
- **Correlation:** Similar to convolution but does not flip the kernel. The kernel slides over the image and computes the similarity at each pixel.

Key Difference: The main difference is that in convolution, the kernel is flipped before the operation, while in correlation, the kernel is not flipped.

Mathematical Definition of Correlation

- **Formula:**

$$g(x, y) = \sum_{i=-k}^k \sum_{j=-k}^k f(x + i, y + j) \cdot h(i, j)$$

Where:

- $f(x, y)$ is the input image.
- $h(i, j)$ is the kernel.
- **Important:** The kernel is not flipped in correlation.